AI & Cyber Individual Project Report

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I have worked on the deep learning part of the project and brought the final results together. I used the MLP model, compared its results with the other models, checked what happened when message-length features were removed, and helped shape the final discussion around limitations.

I also helped organise the final project summary so that the report did not only show high scores, but also explained what those scores actually meant.

Why I Used an MLP

I chose a small Multilayer Perceptron (MLP) because the final dataset was tabular. Each row represented a central-node/contact-node pair, not a message sequence. Because of that, models such as LSTM, GRU, or Transformers were not the best fit for this project.

A larger deep learning model might have looked more advanced, but it would not have matched the structure of the data. The MLP was a better choice because it gave the project a deep learning model while still staying suitable for a small pair-level dataset.

MLP Results

Table 1: MLP Experiment Results

Experiment	Top-1	MRR	F1	Interpretation
MLP base	1.00	1.00	1.00	Perfect score on the base features.
MLP contextual	1.00	1.00	1.00	Perfect score on the enhanced features.
MLP no length	0.90	0.91	0.90	Small drop after removing length features.
SVM no length	1.00	1.00	1.00	Best robustness result in the ablation.

The MLP performed very well on the base and contextual feature sets. It ranked the true pair first in every held-out scenario. But the no-length result was more useful for understanding the model properly. When message-length features were removed, the MLP dropped from 1.00 to 0.90 Top-1 Accuracy. This showed that the model was strong, but still partly affected by length-related signals.

Error Analysis

Table 2: No-Length Error Analysis

Model	True pairs	Missed Top-1	Worst true rank
Random Forest	10	3	5
SVM	10	0	1
MLP	10	1	9

The error analysis helped me understand the results more carefully. The MLP missed one true pair in the no-length setting. This happened in Dataset-3, scenario perp_4, where the true pair was ranked ninth.

This was an important result because it showed that perfect full-feature performance should not be accepted without question. The model worked very well on PASYDA, but the ablation showed that some of the success came from message-length cues.

Deliverables

Table 3: Deep Learning Deliverables

Deliverable	Purpose
MLP base experiment	Tested the MLP on the base pair-level features
MLP contextual experiment	Tested the MLP on the enhanced feature table
MLP no-length experiment	Checked whether the MLP depended on message-length features
tuning_selected_configs_by_outer_fold.csv	Recorded selected model settings during grouped validation
no_length_error_cases.csv	Helped identify concrete missed cases after ablation

My contribution was to show how deep learning could be used in a sensible way for this dataset. I did not choose a complex model just for the sake of it. The MLP matched the data structure better because the project used pair-level tabular features.

The MLP results were strong, but the no-length error case showed that the model was not perfect once an important signal was removed. This helped make the conclusion more honest and more realistic.

I realized that at first it might seem natural to use a bigger deep learning architecture, but the data did not support that. The labels were pair-level, and the final feature table was tabular, so a small MLP was the better fit.

Future Work

I would like to test the MLP on a larger external dataset in future and add calibration analysis. I would also compare ranking-based triage with threshold-based alerts, because real-world use would need careful control of false positives and false negatives.